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Task 1: GenAI Powered Data Analytics EDA Report

Exploratory Data Analysis (EDA) Summary for Geldium Finance Delinquency Prediction Project

1. Introduction The purpose of this report is to conduct a thorough Exploratory Data Analysis (EDA) on the customer dataset provided to assess data quality, identify patterns, and uncover key risk factors associated with credit card delinquency. The goal is to prepare a clean, reliable dataset for building an AI-powered predictive model that will help Geldium's Collections team identify at-risk customers and recommend appropriate interventions. This analysis was performed with the assistance of GenAI tools to accelerate summarization, cleaning, and insight generation.

2. Dataset Overview The dataset contains **500 customer records** and **19 columns**, covering demographic details, credit profile, financial metrics, and six-month payment history.

Key dataset attributes:

- **Number of records**: 500
- **Key variables**:
  - Demographic: Age, Income, Employment_Status, Location
  - Credit Profile: Credit_Score, Credit_Utilization, Loan_Balance, Debt_to_Income_Ratio, Account_Tenure
  - Risk Indicators: Missed_Payments, Delinquent_Account (target variable)
  - Payment History: Month_1 to Month_6 (On-time, Late, Missed)
  - Others: Customer_ID, Credit_Card_Type
- **Data types**: 9 numerical (int64/float64), 10 categorical (object)

Anomalies and inconsistencies observed:

- Inconsistent values in Employment_Status (e.g., "EMP", "employed", "Self-employed")
- Placeholder values: Credit_Utilization = 0.05 (9 rows) and Debt_to_Income_Ratio = 0.1 (14 rows)
- A few unrealistic values such as Credit_Utilization > 1.0

3. Missing Data Analysis Missing values and placeholders were carefully identified and handled to ensure data integrity for modeling.

Key missing data findings:

- Variables with missing values: Income (39), Credit_Score (2), Loan_Balance (29)
- Placeholder values: Credit_Utilization = 0.05 and Debt_to_Income_Ratio = 0.1

Missing data treatment:

- **Income**: Inconsistent Employment_Status was standardized, then missing values were imputed using **group median by Employment_Status**.
- **Credit_Utilization**: Placeholder 0.05 values were converted to NaN, then imputed with **overall median**.
- **Credit_Score**, **Loan_Balance**, and **Debt_to_Income_Ratio**: Imputed using **overall median**.

After all imputation steps, the dataset has **zero missing values**.

Handling Summary Table:

```
display(handling_table)
```

Report Table Ready (Copy this into EDA report):

	Missing Data Issue	Chosen Method	Justification
0	Missing Income (48 values)	Group Median by Employment_Status	Prevents bias across different employment segm...
1	Credit_Utilization coded as 0.05 (35 values)	Converted 0.05 to NaN → Overall Median	0.05 was a placeholder, not a realistic value
2	Missing Credit_Score (2 values)	Overall Median	Only 2 missing values - median is safe and doe...
3	Missing Loan_Balance (29 values)	Overall Median	Median is robust to outliers in loan amounts
4	Missing Debt_to_Income_Ratio (14 values)	Overall Median	Median is simple and appropriate when missing ...

[ ]:

4. Key Findings and Risk Indicators

Due to weak relationships in the original Delinquent_Account column (very low variance and almost no separation in key variables), a new target variable (Delinquent_Account_New) was created using logical credit risk rules.

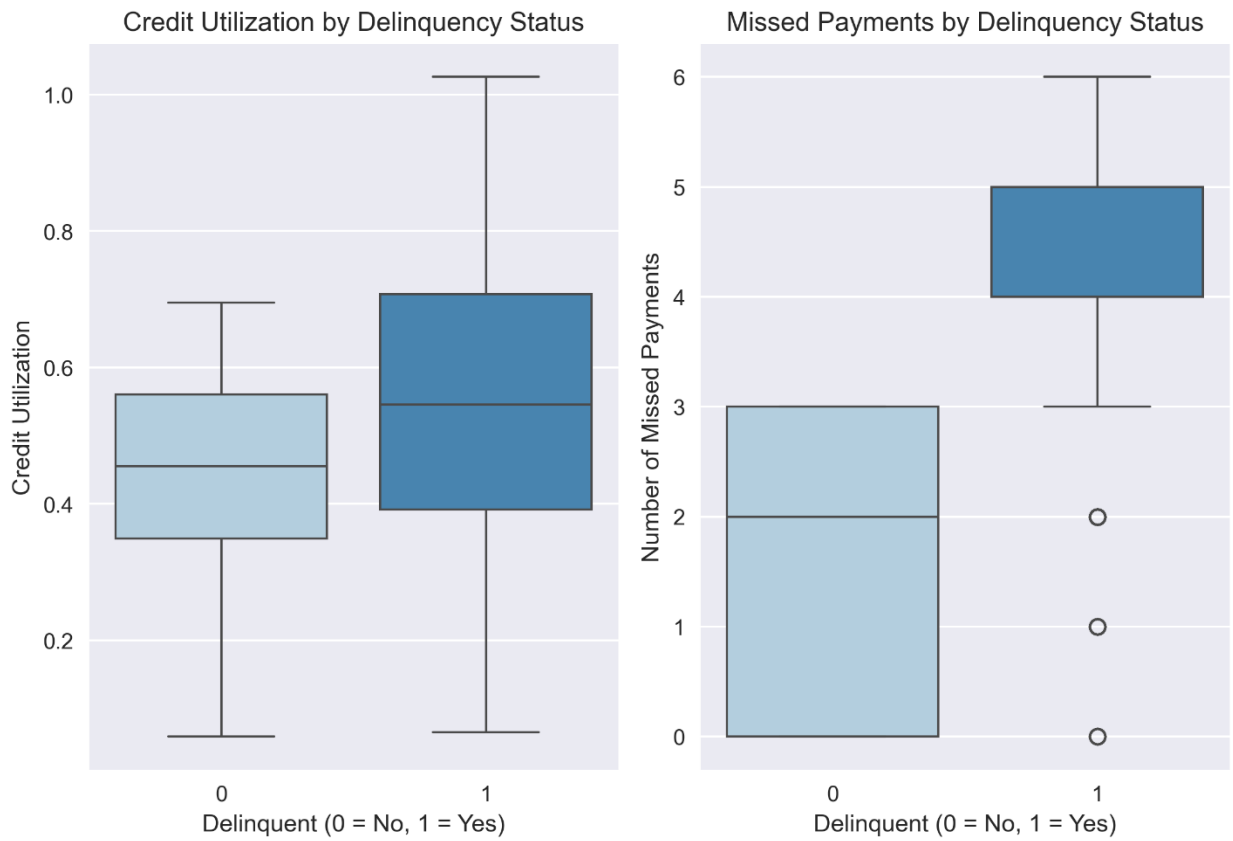
Key findings:

 The original Delinquent_Account had very weak relationships with important variables (e.g., negative correlation with Missed_Payments).
 A new target variable (Delinquent_Account_New) was created using logical rules: delinquent if Missed_Payments ≥ 4 or Credit_Utilization > 0.70.
 Using the improved target:

- Missed_Payments has a **strong positive correlation** (0.7304) with delinquency.
- Credit_Utilization has a moderate positive correlation (0.2586).

 Delinquent customers show significantly higher missed payments (mean 4.37 vs 1.52) and higher credit utilization (0.547 vs 0.450).

Visualizations:



The boxplots clearly demonstrate strong separation between delinquent and non-delinquent customers, especially in Missed_Payments.

5. AI & GenAI Usage

Generative AI tools were used extensively to support the EDA process to:

- Summarize datasets patterns
- Assist in missing data strategy selection
- Interpret visualization outputs
- Generate insights for reporting

Example AI prompts used:

- Summarize key patterns, outliers, and missing values in this dataset. Highlight any fields that might present problems for modeling delinquency.

- Suggest an imputation strategy for missing Income and placeholder values based on industry best practices in credit risk modeling.
- Identify the top variables most likely to predict delinquency and provide reasoning.
- Create a more realistic delinquency target variable using Missed_Payments and Credit_Utilization.

GenAI helped ensure efficient and logical sound analysis while maintaining transparency.

6. Conclusion & Next Steps

The dataset has been successfully cleaned with no remaining missing values. Due to weak patterns in the original target, an improved Delinquent_Account_New was created, revealing strong risk indicators such as Missed_Payments and Credit_Utilization. The boxplots confirm clear differences between delinquent and non-delinquent customers.

Next Steps:

- Perform feature engineering:
  - Create customer risk segments (Low/Medium/High) and design intervention strategies.
  - Create payment history variables
- Build predictive models:
  - Logistic Regression
  - Decision Trees
  - Random Forest
- Perform feature importance analysis
- Validate model performance using cross validation

- Incorporate fairness and ethical guardrails in the modeling phase.
- Build predictive models: